Homework 1

MATH 441: DIFFERENTIAL EQUATIONS

NAME: DATE:

(Exercises are taken from *Elementary Differential Equations and Boundary Value Problems* by William E. Boyce and Richard C. DiPrima)

1. Exercise §2.1 #17. find the solution of the given initial value problem.

$$y' - 2y = e^{2t}, \qquad y(0) = 2$$

2. Exercise §2.1 #19. find the solution of the given initial value problem.

$$t^3y' + 4t^2y = e^{-t}, \qquad y(-1) = 0, \qquad t < 0$$

3. Exercise §2.1 #40. Use the method of variation of parameters to solve the given differential equation

$$y' + (1/t)y = 3\cos 2t, \qquad t > 0$$

4. Exercise §2.2 #7. Solve the given differential equation

$$\frac{dy}{dx} = \frac{x - e^{-x}}{y + e^{y}}$$

5. Exercise §2.2 #22. Solve the intial value problem

$$y' = 3x^2/(3y^2 - 4), \qquad y(1) = 0$$

and determine the interval in which the solution is valid.
Hint: To find the interval of definition, look for points where the integral curve has a vertical tangent.

6. Exercise §2.2 #33.

$$\frac{dy}{dx} = \frac{4y - 3x}{2x - y}$$

(a) Show that the given equation is homogeneous.

(b) Solve the differential equation

(c) Draw a direction field and some integral curves. Are they symmetric with respect to the origin?

7. Exercise §2.6 #10. Determine whether the equation is exact. If it is exact, find the solution.

$$(y/x + 6x) + (\ln x - 2)y' = 0, \qquad x > 0$$